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ID: 12

Project (1)

Spartan6 - DSP48A1

RTL CODE:

Top module

```
module DSP (A, B, C, D, clk, carryin, opmode, Bcin, rstA, rstB, rstM, rstP, rstC,
rstD, rstcarryin, rstopmode, CEA, CEB, CEM, CEP, CEC, CED, CEcarryin, CEopmode,
Pcin, Bcout, Pcout, P, M, carryout, carryoutf);

    parameter A0REG = 0 , A1REG = 1 , B0REG = 0, B1REG = 1;
    parameter CREG = 1, DREG = 1, MREG = 1,PREG = 1, CARRYINREG = 1,CARRYOUTREG =
1,OPMODEREG = 1;
    parameter CARRYINSEL = "OPMODE5", B_INPUT = "DIRECT", RSTTYPE = "SYNC";

    input [17:0] A, B, D, Bcin;
    input [47:0] C, Pcin;
    input [7:0] opmode;
    input clk, carryin, rstA, rstB, rstM, rstP, rstC, rstD, rstcarryin,
rstopmode, CEA, CEB, CEM, CEP, CEC, CED, CEcarryin, CEopmode;

    output [17:0] Bcout;
    output [47:0] P, Pcout;
    output [35:0] M;
    output carryout, carryoutf;

    wire [17:0] A1_wire, B1_wire, A2_wire, B3_wire, D_wire;
    reg [17:0] Pre_add_sub, B0_reg, B2_reg;
    wire [47:0] C_wire, P1_wire, concaten_wire;
    reg [47:0] X_reg, Z_reg, P0_reg;
    wire [35:0] M0_wire, M1_wire;
    wire CIN;
    reg CYI_reg, Cout;
    wire [7:0] opmode_wire;

    REG_MUX #(.REG_EN(OPMODEREG), .RSTTYPE(RSTTYPE), .WIDTH(8)) opmodereg (clk,
rstopmode, CEopmode, opmode, opmode_wire);
    REG_MUX #(.REG_EN(DREG), .RSTTYPE(RSTTYPE), .WIDTH(18)) Dreg (clk, rstD, CED,
D, D_wire);

    //B_INPUT
    always @ (*) begin
```

```
        if (B_INPUT == "DIRECT")
            B0_reg = B;
        else if (B_INPUT == "CASCADE")
            B0_reg = Bcin;
    end

    REG_MUX #(.REG_EN(B0REG), .RSTTYPE(RSTTYPE), .WIDTH(18)) B0reg (clk, rstB,
CEB, B0_reg, B1_wire);


    //pre addition and subtraction
    always @ (*) begin
        if (opmode_wire[6])
            Pre_add_sub = D_wire - B1_wire;
        else
            Pre_add_sub = D_wire + B1_wire;
    end

    //B1_reg_Input
    always @ (*) begin
        if (opmode_wire[4])
            B2_reg = Pre_add_sub;
        else
            B2_reg = B1_wire;
    end

    REG_MUX #(.REG_EN(B1REG), .RSTTYPE(RSTTYPE), .WIDTH(18)) B1reg (clk, rstB,
CEB, B2_reg, B3_wire);

    REG_MUX #(.REG_EN(A0REG), .RSTTYPE(RSTTYPE), .WIDTH(18)) A0reg (clk, rstA,
CEA, A, A1_wire);

    REG_MUX #(.REG_EN(A1REG), .RSTTYPE(RSTTYPE), .WIDTH(18)) A1reg (clk, rstA,
CEA, A1_wire, A2_wire);

    assign concaten_wire = {D_wire[11:0], A2_wire, B3_wire};

    assign M0_wire = B3_wire * A2_wire;

    REG_MUX #(.REG_EN(MREG), .RSTTYPE(RSTTYPE), .WIDTH(36)) Mreg (clk, rstM, CEM,
M0_wire, M1_wire);

    //X_INPUT
    always @ (*) begin
        case (opmode_wire[1:0])
```

```
            0: X_reg = 0;
            1: X_reg = {{12{1'b0}},M1_wire};
            2: X_reg = P1_wire;
            3: X_reg = concaten_wire;
        endcase
    end

    REG_MUX #(.REG_EN(CREG), .RSTTYPE(RSTTYPE), .WIDTH(48)) Creg (clk, rstC, CEC,
C, C_wire);

    //Z_INPUT
    always @ (*) begin
        case (opmode_wire[3:2])
            0: Z_reg = 0;
            1: Z_reg = Pcin;
            2: Z_reg = P1_wire;
            3: Z_reg = C_wire;
        endcase
    end

    //carryin cascade
    always @ (*) begin
        case (CARRYINSEL)
            "OPMODE5": CYI_reg = opmode_wire[5];
            "CARRYIN": CYI_reg = carryin;
            default: CYI_reg = 0;
        endcase
    end

    REG_MUX #(.REG_EN(CARRYINREG), .RSTTYPE(RSTTYPE), .WIDTH(1)) CYIreg (clk,
rstcarryin, CEcarryin, CYI_reg, CIN);




    //Post_add_sub
    always @ (*) begin
        if (opmode_wire[7])
            {Cout,P0_reg} = Z_reg - (X_reg + CIN);
        else
           {Cout,P0_reg} = X_reg + Z_reg + CIN;
    end
```

```
    REG_MUX #(.REG_EN(PREG), .RSTTYPE(RSTTYPE), .WIDTH(48)) Preg (clk, rstP, CEP,
P0_reg, P1_wire);

    REG_MUX #(.REG_EN(CARRYOUTREG), .RSTTYPE(RSTTYPE), .WIDTH(1)) CYOreg (clk,
rstcarryin, CEcarryin, Cout, carryout);

    assign carryoutf = carryout;

    assign Bcout = B3_wire;

    assign P = P1_wire;

    assign Pcout = P1_wire;

    assign M = M1_wire;
endmodule
```

REG_MUX:

```
module REG_MUX (clk, rst, CE, din, dout);

    parameter REG_EN = 1, RSTTYPE = "SYNC", WIDTH = 18;
    input [WIDTH-1:0] din;
    input clk, rst, CE;
    output [WIDTH-1:0] dout;
    reg  [WIDTH-1:0] dreg;

    generate
        if (RSTTYPE == "SYNC") begin
            always @ (posedge clk) begin
                if (rst)
                    dreg <= 0;
                else if (CE == 1)
                    dreg <= din;
            end
        end
        else if (RSTTYPE == "ASYNC") begin
            always @ (posedge clk or posedge rst) begin
                if (rst)
                    dreg <= 0;
                else if (CE == 1)
                    dreg <= din;
            end
```

```
        end
    endgenerate

    assign dout = (REG_EN == 1 ) ? dreg : din;
endmodule
```

TESTBENCH CODE:

```
module DSP_tb ();
    parameter A0REG = 0 , A1REG = 1 , B0REG = 0, B1REG = 1;
    parameter CREG = 1, DREG = 1, MREG = 1,PREG = 1, CARRYINREG = 1,CARRYOUTREG =
1,OPMODEREG = 1;
    parameter CARRYINSEL = "OPMODE5", B_INPUT = "DIRECT", RSTTYPE = "SYNC";

    reg [17:0] A, B, D, Bcin;
    reg [47:0] C, Pcin;
    reg [7:0] opmode;
    reg clk, carryin, rstA, rstB, rstM, rstP, rstC, rstD, rstcarryin, rstopmode,
CEA, CEB, CEM, CEP, CEC, CED, CEcarryin, CEopmode;

    wire [17:0] Bcout;
    wire [47:0] P, Pcout;
    wire [35:0] M;
    wire carryout, carryoutf;

    DSP #(.A0REG (A0REG), .A1REG(A1REG), .B0REG(B0REG), .B1REG(B1REG),
.CREG(CREG), .DREG(DREG), .MREG(MREG), .PREG(PREG), .CARRYINREG(CARRYINREG),
.CARRYOUTREG(CARRYOUTREG), .OPMODEREG(OPMODEREG), .CARRYINSEL(CARRYINSEL),
.B_INPUT(B_INPUT), .RSTTYPE(RSTTYPE) )
    DUT (A, B, C, D, clk, carryin, opmode, Bcin, rstA, rstB, rstM, rstP, rstC,
rstD, rstcarryin, rstopmode, CEA, CEB, CEM, CEP, CEC, CED, CEcarryin, CEopmode,
Pcin, Bcout, Pcout, P, M, carryout, carryoutf);


    //clk
    initial begin
        clk =0;
```

```
        forever
            #1 clk = ~clk;
    end

    initial begin
        rstA = 1;
        rstB = 1;
        rstM = 1;
        rstP = 1;
        rstC = 1;
        rstD = 1;
        rstcarryin = 1;
        rstopmode = 1;

        repeat (10) begin
            A = $random;
            B = $random;
            D = $random;
            Bcin = $random;
            C = $random;
            Pcin = $random;
            opmode = $random;
            carryin = $random;
            CEA = $random;
            CEB = $random;
            CEM = $random;
            CEP = $random;
            CEC = $random;
            CED = $random;
            CEcarryin = $random;
            CEopmode = $random;
            @ (negedge clk);

            if ((Bcout != 0) || (P != 0) || (Pcout != 0) || (M != 0) || (carryout
!= 0) || (carryoutf != 0)) begin
                $display ("Error in reset");
                $stop;
            end
        end

        rstA = 0;
        rstB = 0;
        rstM = 0;
        rstP = 0;
        rstC = 0;
```

```
        rstD = 0;
        rstcarryin = 0;
        rstopmode = 0;

        CEA = 1;
        CEB = 1;
        CEM = 1;
        CEP = 1;
        CEC = 1;
        CED = 1;
        CEcarryin = 1;
        CEopmode = 1;

        //PATH 1

            opmode = 8'b11011101;
            A = 20;
            B = 10;
            C = 350;
            D = 25;

            repeat (10) begin
                Bcin = $random;
                Pcin = $random;
                carryin = $random;
                @ (negedge clk);
                @ (negedge clk);
                @ (negedge clk);
                @ (negedge clk);
                if ((Bcout != 'hf) || (P != 'h32) || (Pcout != 'h32) || (M != 
'h12c) || (carryout != 0) || (carryoutf != 0)) begin
                    $display ("Error in PATH 1");
                    $stop;
                end
            end

        // PATH 2

            opmode = 8'b00010000;
            A = 20;
            B = 10;
            C = 350;
            D = 25;

            repeat (10) begin
```

```
                Bcin = $random;
                Pcin = $random;
                carryin = $random;
                @ (negedge clk);
                @ (negedge clk);
                @ (negedge clk);
                if ((Bcout != 'h23 ) || (P != 0) || (Pcout != 0) || (M != 'h2bc)
|| (carryout != 0) || (carryoutf != 0)) begin
                    $display ("Error in PATH 2");
                    $stop;
                end
            end

        //PATH 3
            opmode = 8'b00001010;
            A = 20;
            B = 10;
            C = 350;
            D = 25;

            repeat (10) begin
                Bcin = $random;
                Pcin = $random;
                carryin = $random;
                @ (negedge clk);
                @ (negedge clk);
                @ (negedge clk);
                if ((Bcout != 'ha ) || (P != 0) || (Pcout != 0) || (M != 'hc8) ||
(carryout != 0) || (carryoutf != 0)) begin
                    $display ("Error in PATH 3");
                    $stop;
                end
            end

        //PATH4
            opmode = 8'b10100111;
            A = 5;
            B = 6;
            C = 350;
            D = 25;
            Pcin = 3000;

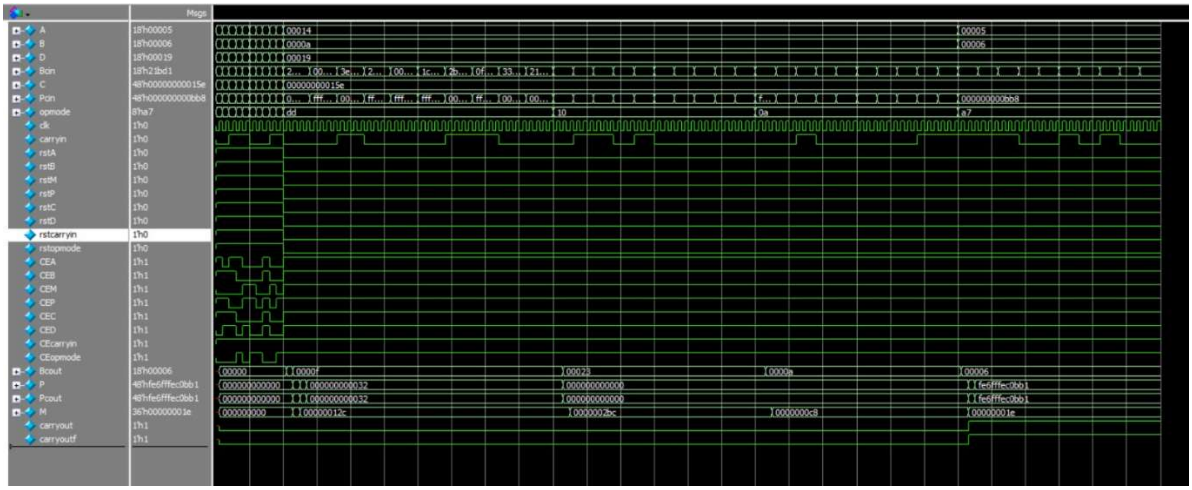
            repeat (10) begin
                Bcin = $random;
                carryin = $random;
```

```
                @ (negedge clk);
                @ (negedge clk);
                @ (negedge clk);
                if ((Bcout != 'h6 ) || (P != 'hfe6fffec0bb1) || (Pcout !=
'hfe6fffec0bb1) || (M != 'h1e) || (carryout != 1) || (carryoutf != 1)) begin
                    $display ("Error in PATH 4");
                    $stop;
                end
            end
        $stop;
    end
endmodule
```

DO FILE:

```
vlib work
vlog DSP.v DSP_tb.v, REG_MUX.v
vsim -voptargs=+acc work.DSP_tb
add wave *
run -all
#quit -sim
```

QuestaSim Snippet:



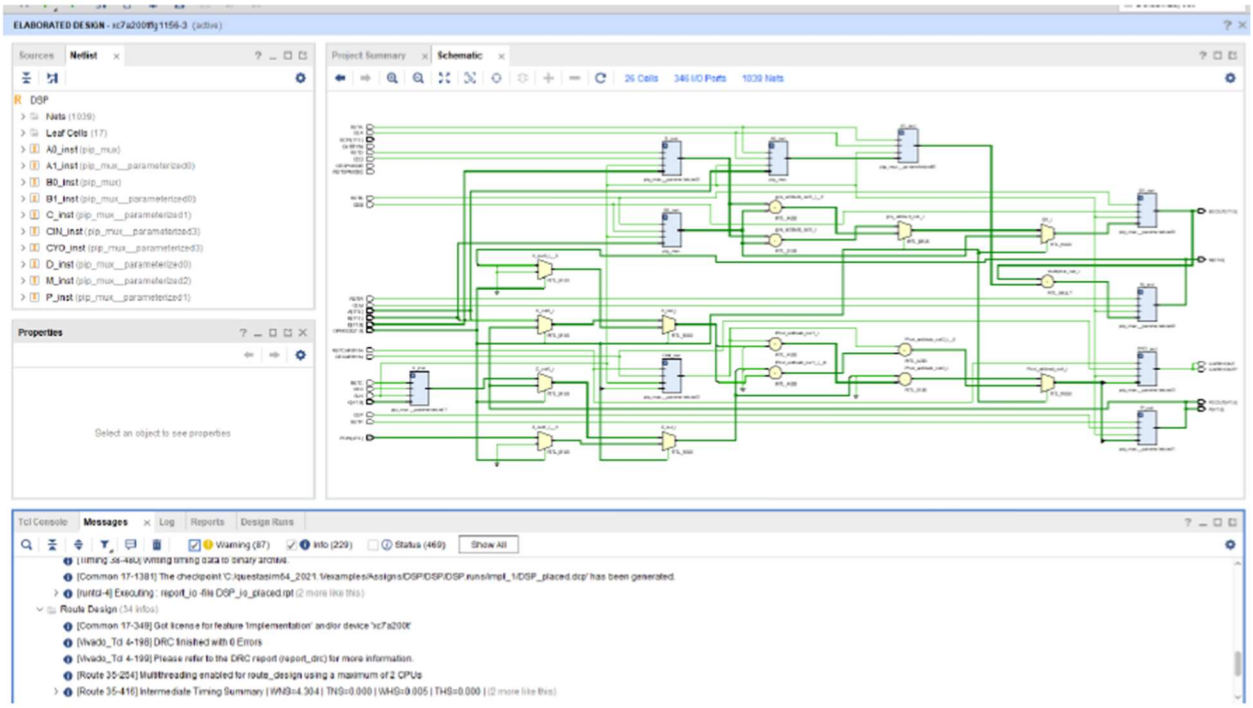
Constraint File:

```
create_clock -period 10.000 -name CLK -waveform {0.000 5.000} [get_ports CLK]

set_property PACKAGE_PIN W5 [get_ports CLK]

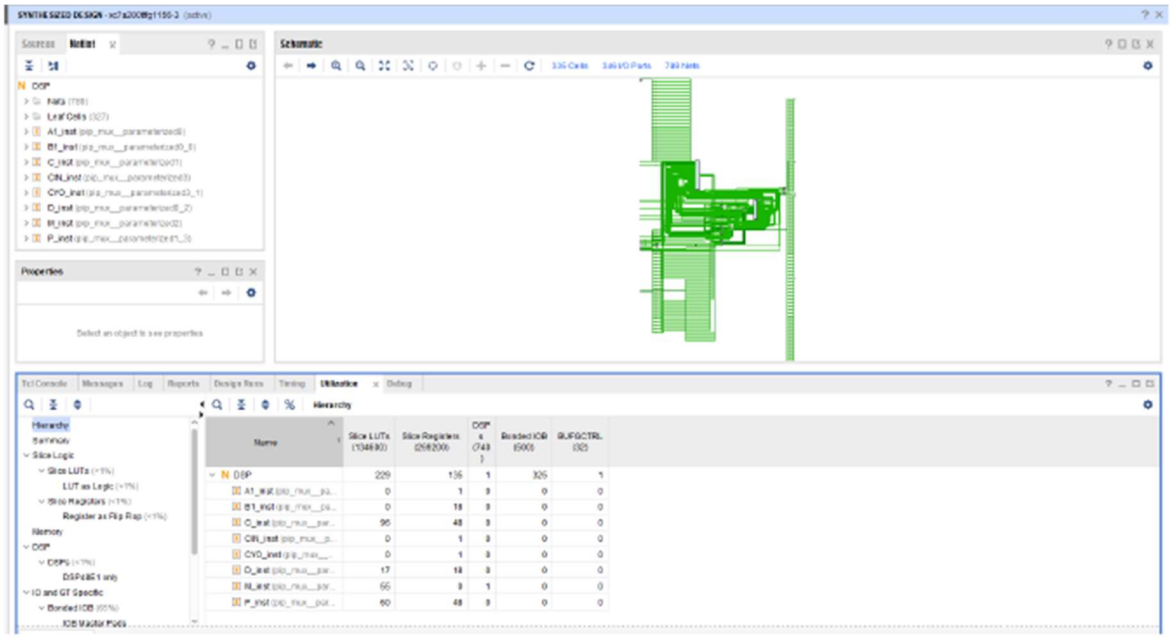
set_property IOSTANDARD LVCMOS33 [get_ports CLK]
```

Elaboration:

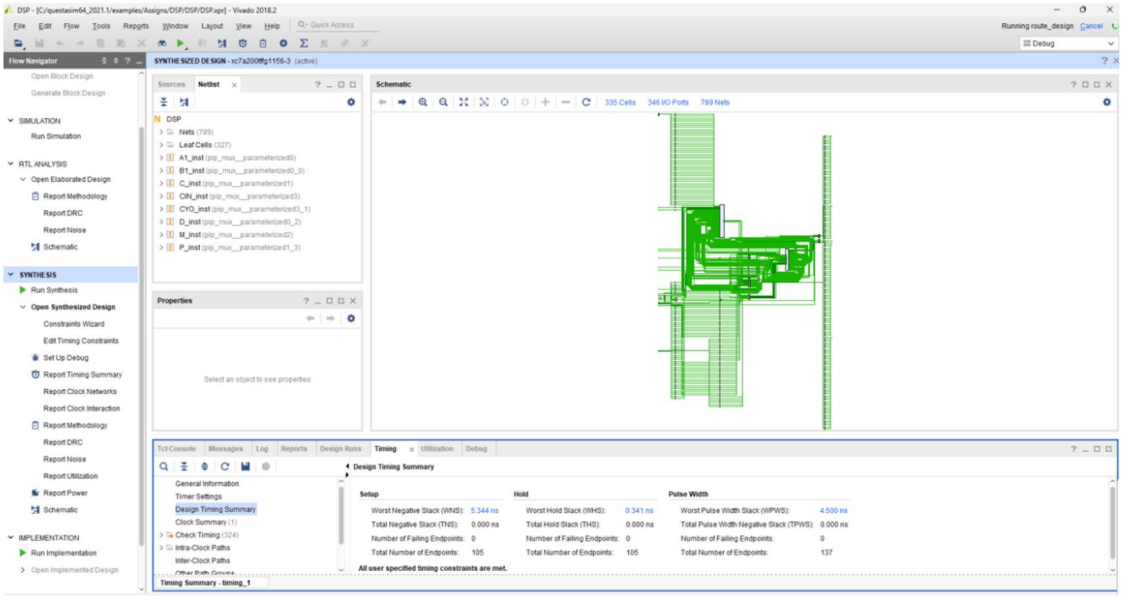


Synthesis:

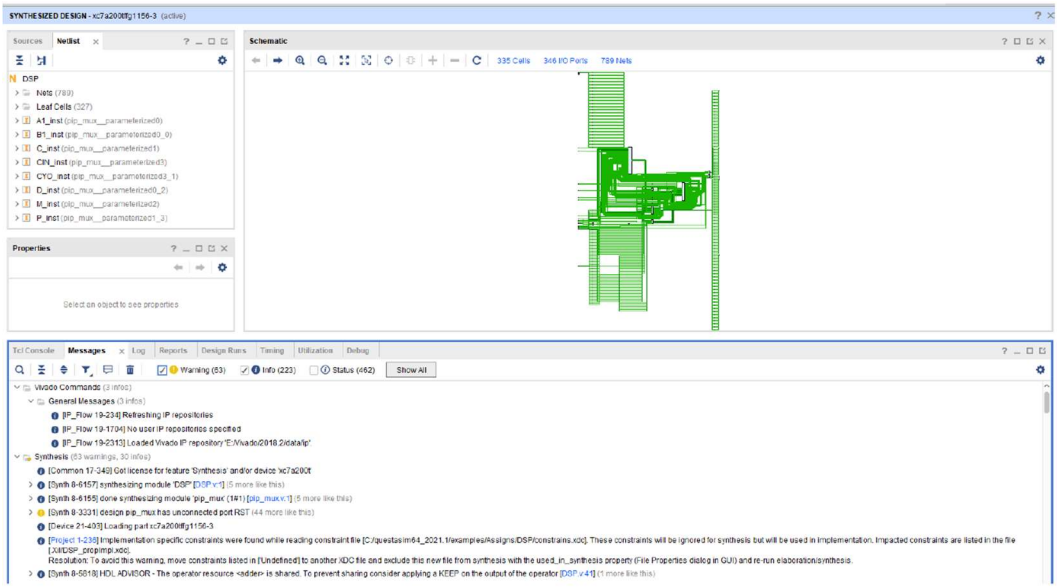
1. Schematic and Utilization:



2. Timing Report

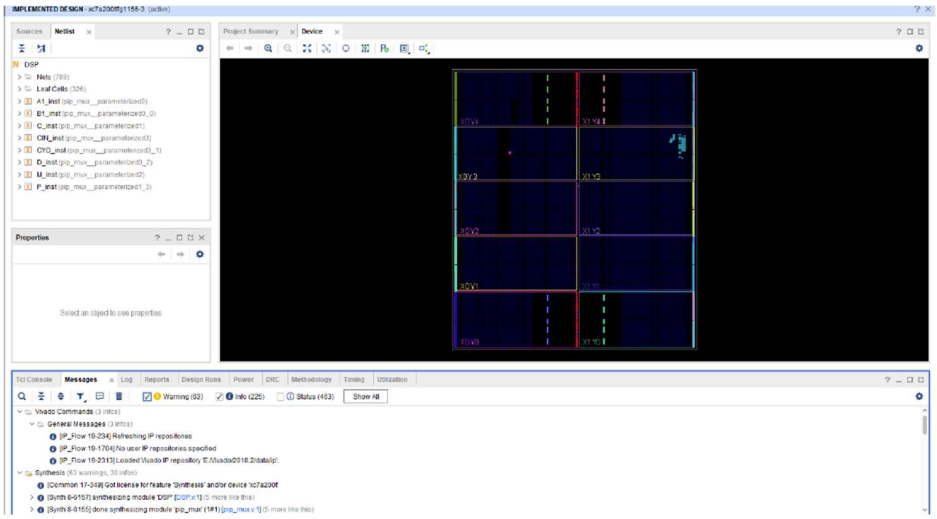


3. Messages:

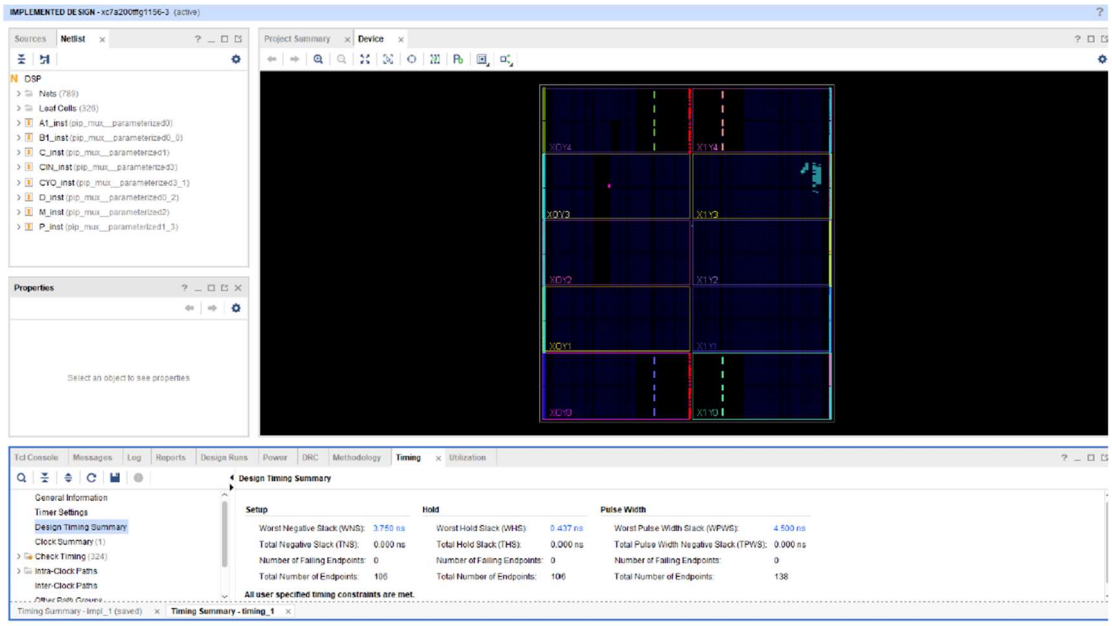


Implementation:

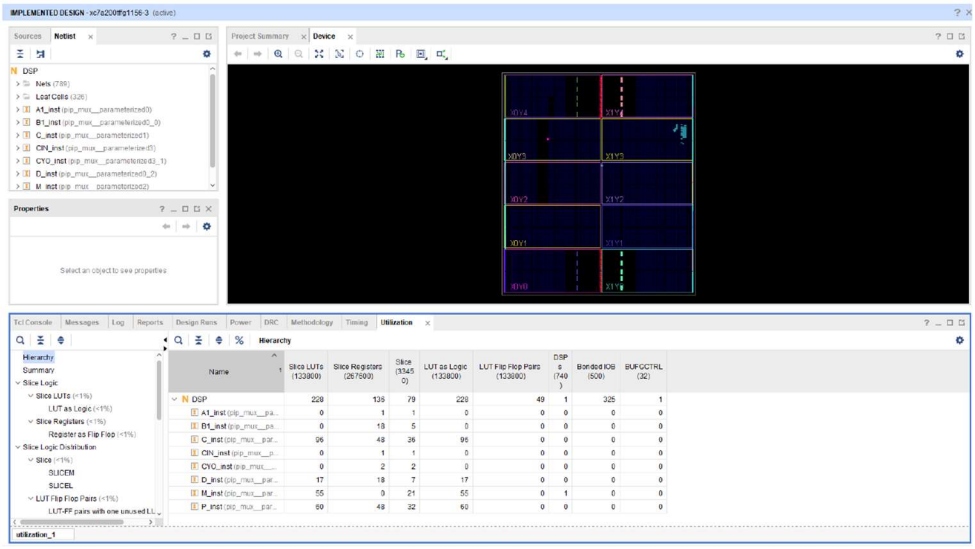
1. Device View, Netlist and Messages:



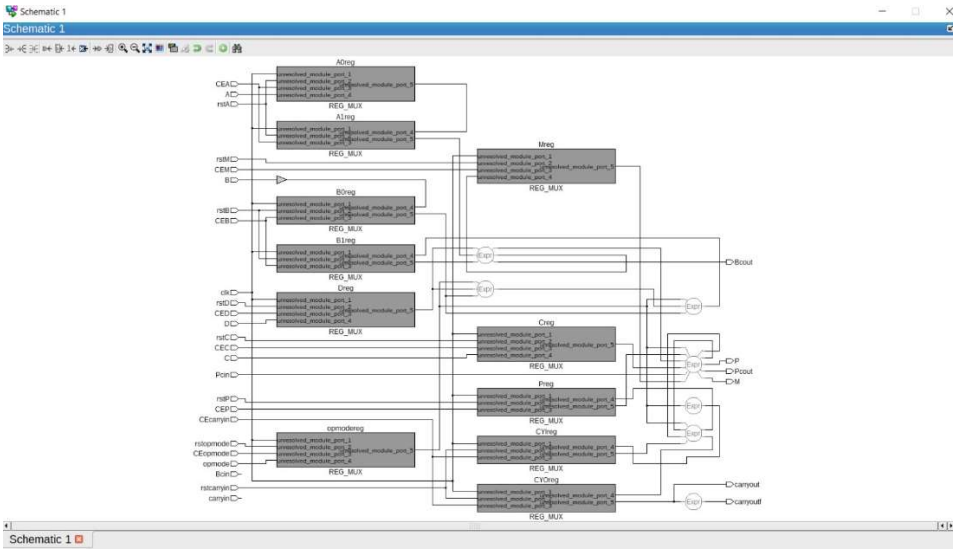
2. Timing Report:

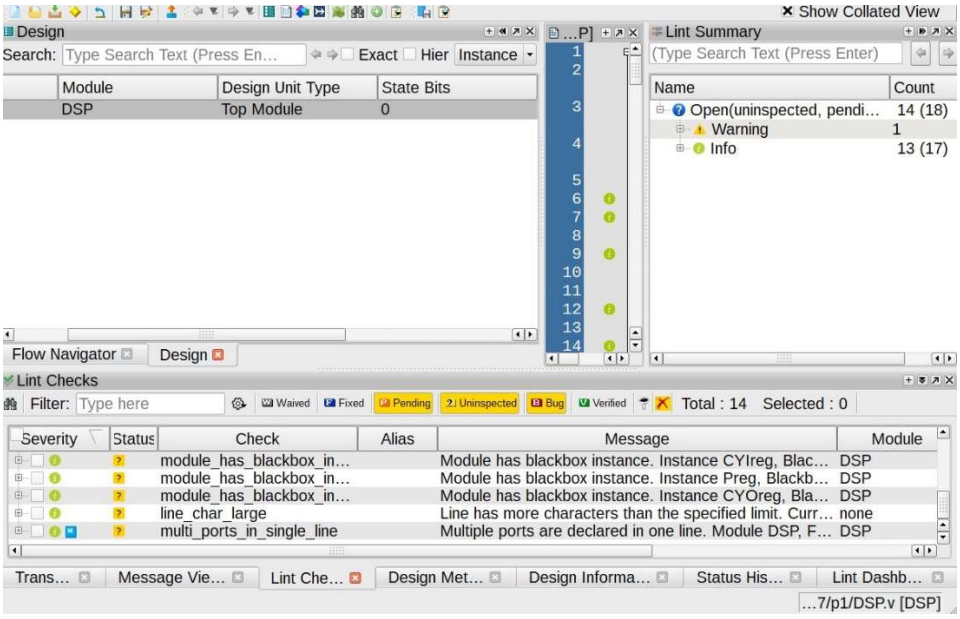


3. Utilization Report:



Linting:


Schematic 1
Schematic 1
AOreg
A1reg
Mreg
BOreg
B1reg
Dreg
Creg
Preg
CYIreg
CYOreg
opmodereg
REG_MUX
Bcout
P
Pcout
M
carryout
carryoutf
Schematic 1


Show Collated View
Design
Search: Type Search Text (Press En... Exact Hier Instance
Module | Design Unit Type | State Bits
DSP | Top Module | 0
Lint Summary
(Type Search Text (Press Enter)
Name | Count
Open(uninspected, pendi... 14 (18)
Warning 1
Info 13 (17)
Flow Navigator
Design
Lint Checks
Filter: Type here
Waived Fixed Pending Uninspected Bug Verified
Total : 14 Selected : 0
Severity | Status | Check | Alias | Message | Module
module_has_blackbox_in... | Module has blackbox instance. Instance CYIreg, Blac... | DSP
module_has_blackbox_in... | Module has blackbox instance. Instance Preg, Blackb... | DSP
module_has_blackbox_in... | Module has blackbox instance. Instance CYOreg, Bla... | DSP
line_char_large | Line has more characters than the specified limit. Curr... | none
multi_ports_in_single_line | Multiple ports are declared in one line. Module DSP, F... | DSP
Trans... Message Vie... Lint Che... Design Met... Design Informa... Status His... Lint Dashb...
...7/p1/DSP.v [DSP]